Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building **patient-centric, mobile, and web applications**; experienced in React, TypeScript, Python, secure APIs, relational data, testing, debugging, documentation, Agile delivery, and cross-functional product collaboration.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Developed **secure Python services and relational data workflows** across FastAPI, PostgreSQL, SQL Server, and Azure storage, supporting 500+ users with row-level isolation and reducing dashboard latency 71%.
- Maintained high-quality integrations for 5,000+ monthly workflows, adding validation, reconciliation, explicit states, error handling, and failure isolation while documenting system behavior and sustaining 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **React interfaces and Python Flask APIs** for uploads, processing, history, reporting, and feedback, serving 150+ users and 10,000+ daily requests while reducing turnaround time 60%.
- Debugged complex application defects and stability issues across UI, API, file-processing, and model integrations, then shipped tested fixes through sprint planning, code review, retrospectives, and GitHub Actions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Implemented **reusable secure API and data integration components** using Python, Java, SQL, AWS Glue, S3, OOP patterns, and caching, improving response times 30% and pipeline efficiency 68%.
- Collaborated with Microsoft and GitHub stakeholders through Agile sprints, documenting integration workflows, reviewing code, troubleshooting defects, and maintaining 20+ unit and integration tests across three operating systems.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **React and Node.js user experiences** across eight services for authentication, records, permissions, notifications, and reporting, securely synchronizing 2M+ records through role-based access, WebSockets, and Azure SQL.
- Improved application security, quality, and stability using JWT, encrypted Azure SQL, Jest, Selenium, Sentry, Azure Monitor, Docker, and Jenkins, achieving 95%+ build validation and 50% faster deployments.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **responsive mobile-first React Native and TypeScript solutions** across Android, iOS, and web, integrating Flask APIs and Firebase synchronization for 50,000+ records with sub-200-millisecond application responses.
- Partnered closely with customers to translate requirements into end-to-end features, documenting 12+ interfaces and delivering secure authentication, notifications, data workflows, validated webhooks, reporting, and real-time status updates.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, HTML, CSS, C++
Applications: React.js, React Native, Next.js, Node.js, Express.js, FastAPI, Flask, responsive design
Healthcare & APIs: Patient workflows, consent-based access, REST APIs, WebSockets, notifications, interoperability
Data & Security: PostgreSQL, SQL Server, MongoDB, Azure SQL, encryption, authentication, role-based access
Engineering & Delivery: Data structures, algorithms, Git, GitHub Actions, CI/CD, Jest, Selenium, PyTest, code reviews, Agile/Scrum, documentation

Projects

HMIS Live | *React, TypeScript, Healthcare APIs*

- Built a **patient-centric healthcare application** with consent-based history access, provider dashboards, appointments, REST integrations, and notifications, halving record retrieval time while supporting 100+ daily workflows and reducing administrative workload 30%.

Big-Brain Live | *Secure Data, PostgreSQL, REST APIs*

- Developed a **secure document application** processing 50,000+ records with authenticated uploads, access controls, PostgreSQL metadata, REST APIs, incremental indexing, and collaborative workflows, reducing retrieval latency 40% and review time 60%.

IoT Authentication Analysis | *Cryptography, Access Control, Security*

- Designed and evaluated **secure authentication models** across three user scenarios and four permission tiers, documenting integration assumptions and replacing centralized verification with tamper-resistant workflows while reducing repeated key checks 50%.